Bangladesh University of Engineering and Technology

Department of Chemistry

CHEM113: Chemistry

For Students of CSE

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Course Content

CHEM113 : Chemistry

Total Credit Hours: 3 ; L1/T2

Quantum concept in atomic structure, VSEPRT; molecular geometry, Quantum concept in bonding; VBT and MOT, Frontier MOT and electronic transition, Silicon chemistry, Properties of solutions, Colloid and Nano-chemsitry, ~~**Phase rule and phase diagram; Energy and chemistry, Electrochemistry; electrolytic conduction, corrosion**~~**, devices for energy storage, Chemistry of biodegradable and conductive polymer; LED, LCD/touch screen,** Chemistry of proteins, nucleic acids (DNA, RNA), carbohydrates and lipids; Introduction to computational chemistry; Design of new molecules, materials and drug.

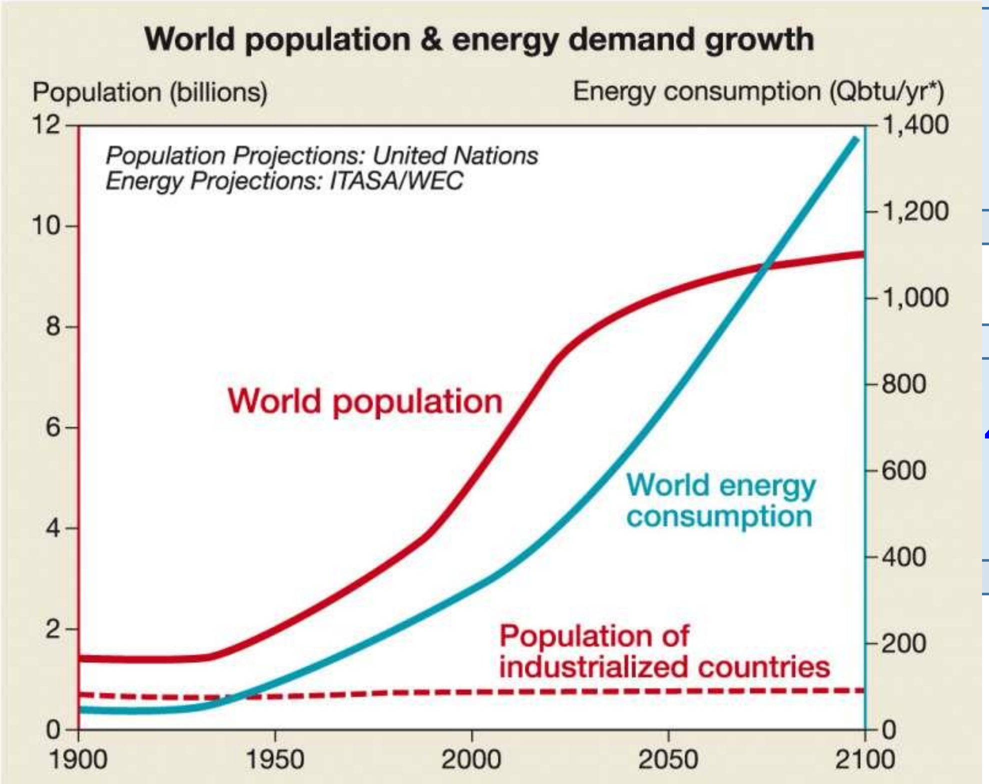
Contents

Energy demand and energy forms_ A short overview

Energy storage_ Importance

Energy storage devices and their type

Background



Distrib

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Background

Special Uses of Electrical Energy

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b

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d

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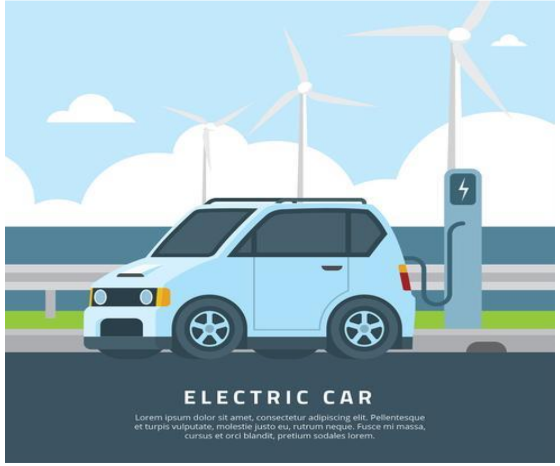

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Requires storage device

Fig. (a-d) Laptop, Tablet, Smart Phone and their illustrative use with dependency, (e) an Electric car is being charged at supercharger station and (f) illustration of electric car at the wind field.

Energy forms

Chemical energy
Thermal energy
Light energy
Electrical energy
Kinetic energy
Wind energy
Mechanical energy
Sound energy

Energy Storage Devices

— Energy Storage —

‘capture of energy produced at one time for use at a later time’

An energy storage device refers to a device used to store energy in various forms such as supercapacitors, batteries, and thermal energy storage systems. It plays a crucial role in ensuring the safety, efficiency, and reliable functioning of microgrids by providing a means to store and release energy as needed.

These devices are crucial for managing energy supply and demand, especially with intermittent renewable energy sources like solar and wind. They can be broadly categorized into mechanical, electrochemical, chemical, electromagnetic, and thermal storage systems.

Energy Storage : Importance

Energy storage is a critical component of a modern, sustainable, and reliable energy system, offering numerous benefits across various sectors and applications.

Energy storage is crucial for a reliable, sustainable, and cost-effective energy system. Its importance is diverse, e.g. it helps integrate renewable energy sources, reduce peak demand, enhance grid stability, and mitigate power outages, making it vital for a smooth energy transition.

- ❑ Integrating Renewable Energy
- ❑ Enhancing Grid Stability and Reliability
- ❑ Improving Energy Efficiency and Reducing Costs
- ❑ Enabling Diverse Applications
- ❑ Reducing Environmental Impact

Energy Storage : Importance

1. Integrating Renewable Energy

Overcoming intermittency: Renewable energy sources like solar and wind are intermittent, meaning their availability fluctuates. Energy storage systems can store excess energy generated during peak production and release it when needed, ensuring a consistent supply of clean energy.

Enabling wider adoption: By smoothing out the variability of renewables, energy storage makes them more reliable and facilitates their integration into the grid, accelerating the transition to a cleaner energy system.

2. Enhancing Grid Stability and Reliability

Reducing peak demand: Energy storage can absorb excess energy during off-peak hours and release it during peak demand, reducing strain on the grid and avoiding the need for expensive infrastructure upgrades.

Providing backup power: In case of grid outages or disruptions, energy storage systems can provide immediate backup power, ensuring critical services remain operational and minimizing downtime.

Supporting grid stability: Energy storage can help regulate voltage and frequency, maintaining the stability and reliability of the power grid.

Energy Storage : Importance

3. Improving Energy Efficiency and Reducing Costs

Optimizing energy consumption: Energy storage allows for time-shifting of energy use, enabling consumers to utilize stored energy during peak hours when electricity prices are higher, reducing their **energy bills**.

Deferring infrastructure investments: By managing demand and optimizing energy flow, storage can help defer or avoid costly investments in new transmission and distribution infrastructure.

4. Enabling Diverse Applications

Microgrids: Energy storage is essential for the development and operation of microgrids, enabling localized energy generation and distribution, particularly in remote areas or during emergencies.

Electric vehicles: Battery storage systems are crucial for the widespread adoption of electric vehicles, providing charging infrastructure and supporting grid stability.

Industrial and commercial facilities: Storage can help businesses optimize their energy consumption, reduce costs, and improve resilience by providing backup power and managing demand.

Energy Storage : Importance

5. Reducing environmental impact

Energy storage enables electricity to be saved for a later, when and where it is most needed. This creates efficiencies and capabilities for the electric grid— including the ability to reduce greenhouse gas (GHG) emissions.

By introducing more flexibility into the grid, energy storage can help integrate more solar, wind and distributed energy resources. It can also improve the efficiency of the grid – increasing the capacity factor of existing resources – and offset the need for building new pollution-emitting peak power plants.

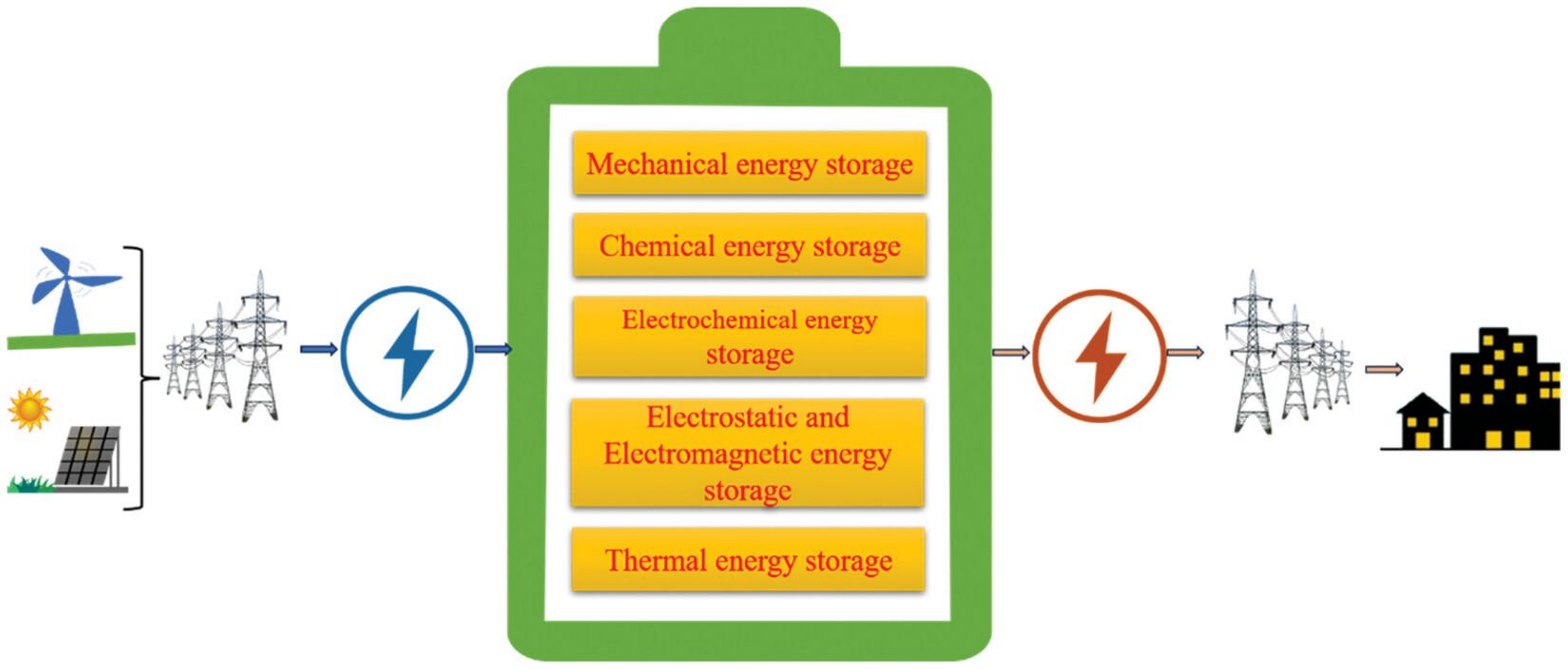
Energy Storage Device_ Types

Energy storage systems can be classified based on several criteria, such as the type of stored energy, the technology employed, their intended application, and their capacity. Energy storage into five main types:

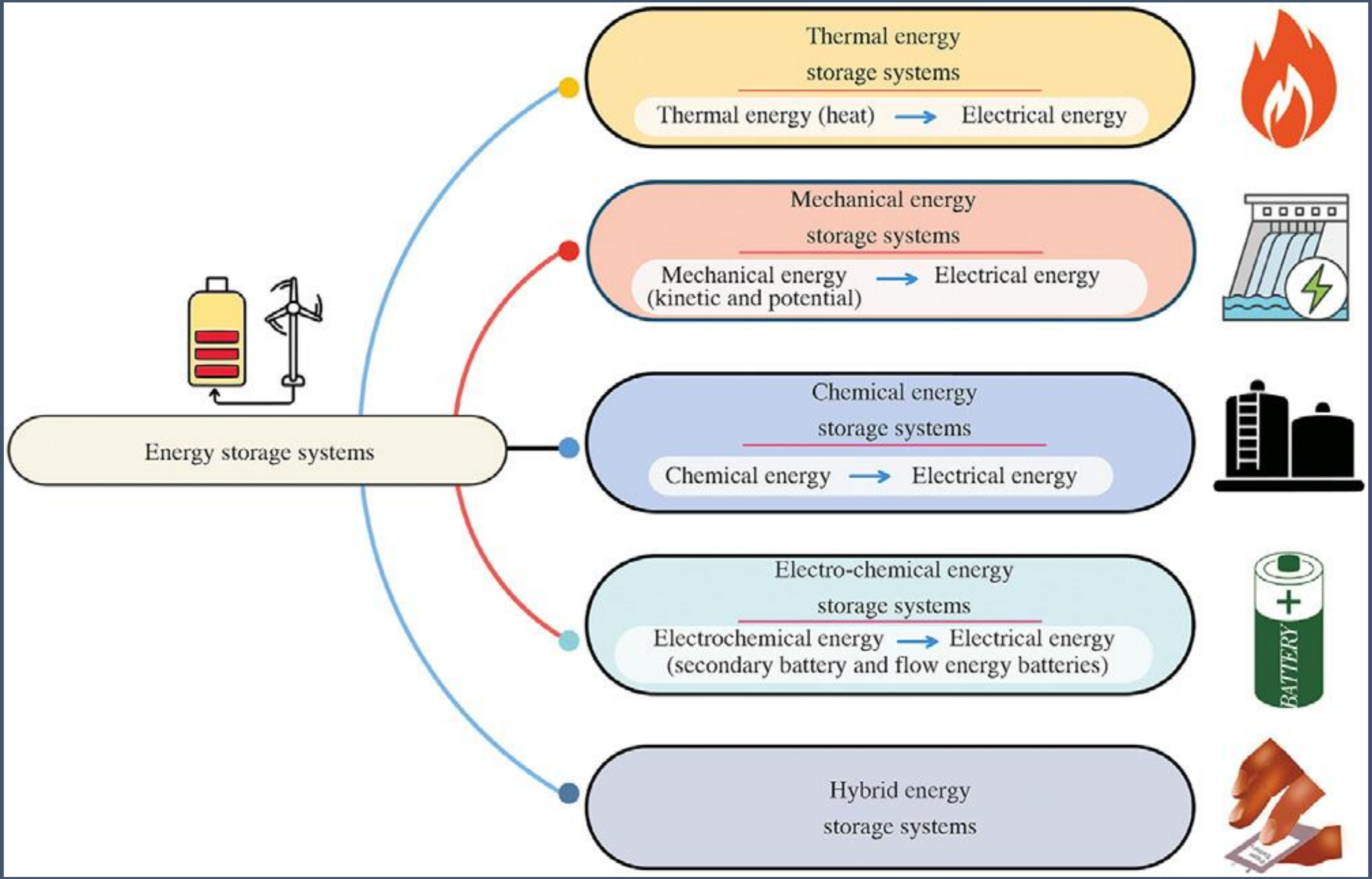
i) mechanical energy storage
ii) chemical energy storage
iii) electrochemical energy storage
iv) electrostatic and electromagnetic energy storage
v) thermal energy storage.

These classifications provide a framework for understanding the diverse ways in which energy can be stored and utilized efficiently. Each type of energy storage has its advantages and limitations, making them suitable for different applications and contexts.

Energy Storage_ Types



Energy Storage_ Types



Energy Storage_ Types

Mechanical

(i) Pumped Hydro: Water is pumped to a higher elevation and released to generate electricity when needed.

(ii) Compressed Air: Air is compressed and stored, then released to drive a turbine.

(iii) Flywheels: A rotating wheel stores kinetic energy.

Electrochemical

(i) Batteries: Convert chemical energy into electrical energy and vice versa.

(ii) Flow Batteries: Electrolyte solutions store energy.

Thermal

(i) Sensible Heat: Stores energy by changing the temperature of a material.

(ii) Latent Heat: Stores energy through phase changes (solid to liquid, liquid to gas).

Thermochemical: Stores energy through chemical reactions.

Energy Storage_ Types

Supercapacitors: Bridge the gap between batteries and capacitors.

Superconducting Magnetic Energy Storage (SMES): Stores energy in a magnetic field.

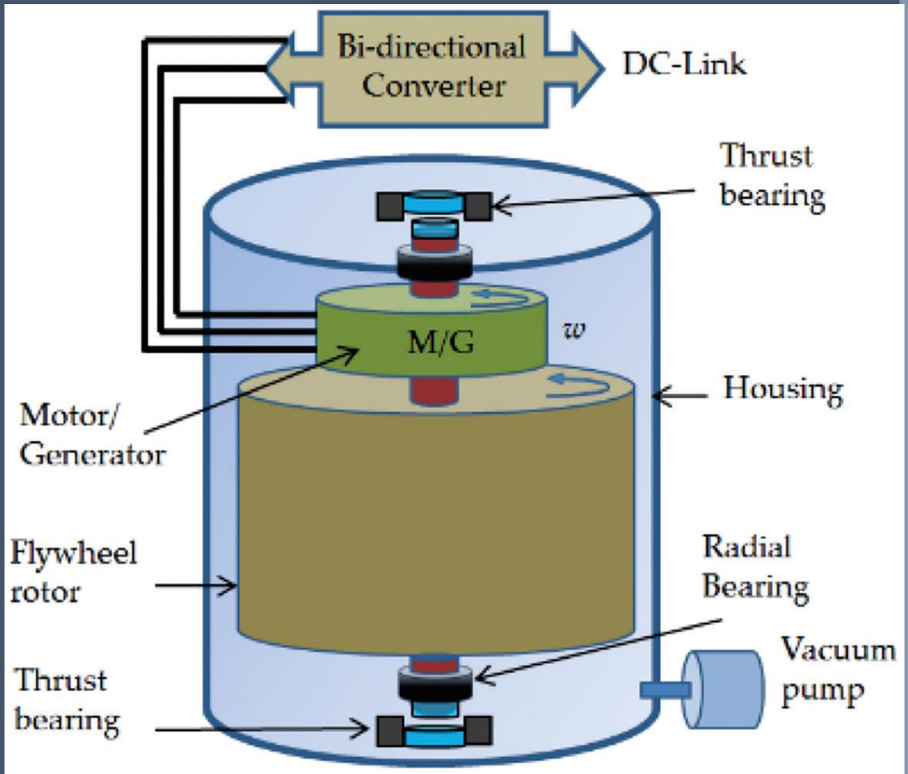
Chemical

Hydrogen Storage: Hydrogen gas is stored and can be used in fuel cells.

Biofuels: Organic matter is stored and converted into energy.

Energy Storage Device_ Flywheel Energy Storage (FES)

Flywheels are an electromechanical arrangement designed to store *kinetic energy* by harnessing the angular momentum of rotating masses such as discs or cylinders. They have recently gained recognition as a viable solution for storing substantial amounts of energy on *a large scale*.

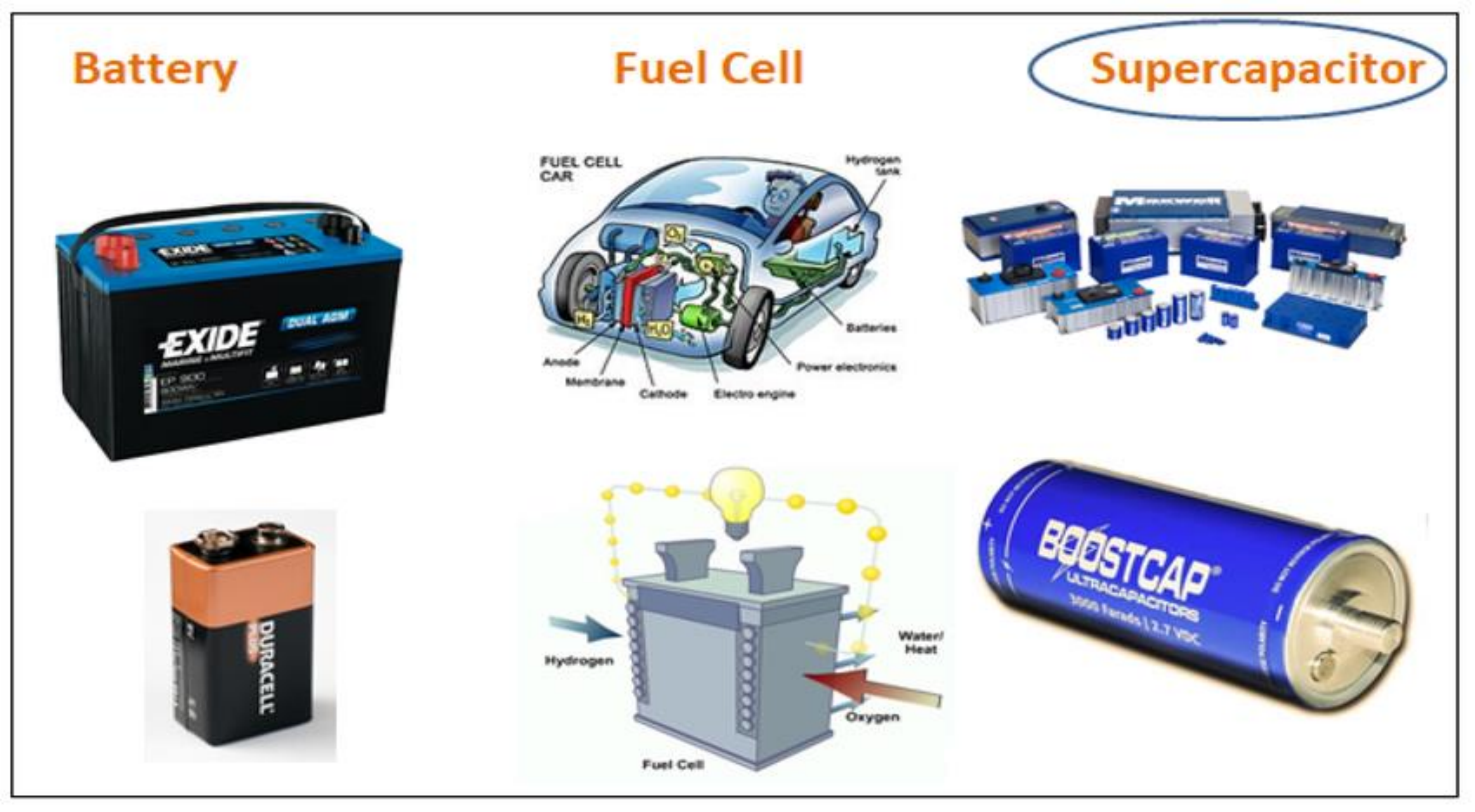


A flywheel operates by accumulating excess energy within the wheel's rotation (spinning mass) and later converts that stored energy, which is accelerated through the assistance of an electric device functioning as a motor. The spinning wheel acts like a storage bank for energy. This process occurs during the charging of a flywheel energy storage system. The same motor switches to being a generator when it gives out energy from the spinning flywheel to the power system.

FES systems come in two types: low-speed and high-speed. Low-speed FES is used in industries and spins around 10,000 times per minute or less. High-speed FES is a bit more complex. This FES stabilizes power grids, provides backup power, and manages renewable energy fluctuations.

Energy Storage Device_ Electrochemical Energy Storage

Electrochemical energy storage system undergoes chemical process to store and produce electricity.



Energy Storage Device_ Battery

Batteries are the most widely used electrochemical energy storage systems in industrial and household applications.

- Two types namely primary and secondary batteries.
- Primary batteries are the non-rechargeable when the chemicals present in them were completely consumed.
- Secondary batteries can be recharged and discharged multiple times.

Primary	Secondary
• Disposable/single-use	• Rechargeable
• Non-rechargeable	• Discharge through electric current
• Internal reactions change batteries irreversibly during discharge	• Reverse current restores electrons to original composition, allowing recharge to happen

Energy Storage Device_ Battery

The rechargeable secondary batteries were widely used in the power system applications. It consists of a cathode, an anode and an electrolyte medium. When the battery is connected to an external circuit, chemical reaction occurs, results in the flow of electrons between the electrodes through the electrolyte and external circuit, and the electromotive force (EMF) is induced. The output voltage produced by these batteries will be less than 2 V. To achieve the required voltage and current, they will be connected together in parallel or series configurations.

Based on the electrode materials and electrolytes used in the system, the secondary batteries were further classified as Lead-acid battery, Nickel-cadmium battery, Sodium-sulfur battery, Lithium-ion battery and flow batteries.